

Database Concepts - Mini Textbook

1. Introduction to Databases & DBMS

A Database is a collection of related data that can be stored, retrieved, and managed efficiently. A Database Management System (DBMS) is software that allows users to define, create, maintain, and control access to databases. Examples: MySQL, Oracle, PostgreSQL, MongoDB.

2. Database Models

Different models define how data is structured and accessed: - Hierarchical Model: Data stored in a tree-like structure. - Network Model: Data represented by records connected by links. - Relational Model: Data stored in tables (relations) with rows (tuples) and columns (attributes). - NoSQL Models: Document, Key-Value, Graph databases designed for unstructured or semi-structured data.

3. Entity-Relationship (ER) Model

The ER model uses entities, attributes, and relationships to represent real-world data. Entities are objects (e.g., Student, Course), attributes describe them (e.g., Name, RollNo), and relationships define associations (e.g., Student ENROLLED_IN Course). ER diagrams visually represent this model.

4. Relational Model & Keys

The Relational Model organizes data into relations (tables). Each table has rows (records) and columns (fields). Keys ensure uniqueness and relationships: - Primary Key: Uniquely identifies each record. - Foreign Key: Refers to the primary key in another table. - Candidate Key: Possible choices for primary key. - Composite Key: Key formed from multiple attributes.

5. Structured Query Language (SQL) Basics

SQL is the standard language to interact with relational databases. Examples: - CREATE TABLE Students (id INT, name VARCHAR(50)); - INSERT INTO Students VALUES (1, 'Alice'); - SELECT * FROM Students; - UPDATE Students SET name='Bob' WHERE id=1; - DELETE FROM Students WHERE id=1;

6. Advanced SQL

SQL supports complex queries: - Joins (INNER, LEFT, RIGHT, FULL) combine data from multiple tables. - Views are virtual tables based on queries. - Indexes speed up searching. - Aggregate functions (COUNT, AVG, SUM, MAX, MIN) summarize data.

7. Normalization & Functional Dependencies

Normalization is the process of organizing data to reduce redundancy and improve integrity. - 1NF: Remove repeating groups. - 2NF: Remove partial dependency. - 3NF: Remove transitive dependency. - BCNF: Stronger version of 3NF. Functional dependency defines relationships between attributes, e.g., StudentID → Name.

8. Transactions & Concurrency Control

A transaction is a sequence of database operations that form a logical unit of work. Transactions follow ACID properties: - Atomicity: All or nothing execution. - Consistency: Database moves from one valid state to another. - Isolation: Transactions execute independently. - Durability: Changes persist after commit. Concurrency control prevents conflicts when multiple users access the database simultaneously.

9. Database Storage & Indexing

Databases use storage techniques like heap files, sequential files, and hashing. Indexing improves query performance by creating data structures (like B-Trees, Hash Indexes) that allow fast searching and retrieval.

10. Database Security & Recent Trends

Database security involves authentication, authorization, encryption, and auditing. Recent trends include: - NoSQL databases for big data. - Cloud databases (AWS RDS, Azure SQL). - Distributed databases. - NewSQL: Combines scalability of NoSQL with relational features.